CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in building features end to end, from Python back-end services and databases to React front ends, with LLMs and agentic coding tools in the daily workflow. Most recently sole developer of an agentic knowledge-extraction system and its expert web app for the German energy sector, working closely with domain experts to turn their questions into clear decisions about what to build.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Developed Expert Interface, a self-serve graph curation web app (React front end, Python microservices back end) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes themselves.
- Sole developer owning the full knowledge-extraction pipeline end-to-end (data ingestion, chunking, prompt engineering, and indexing with agentic, deterministic, and monolithic strategies), reporting directly to the CTO and collaborating with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent extraction algorithms (LangGraph: entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for knowledge extraction from complex regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% via an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Delivered as an AI-assisted developer, using agent-based coding tools (Claude Code, custom skills, agentic coding workflows) for fast development while keeping the code strictly typed, documented, and grounded in clean code principles.
- **Tech stack: Python, FastAPI, React, Postgres, MongoDB, Neo4j, LangGraph, LangChain, LiteLLM, MCP, RAG, GraphRAG, Vector Search, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application using LangGraph to optimize decision-making workflows for vector database queries, general inference, and summarization tasks.
- Worked with business stakeholders to understand their decision-making workflows before designing the solution; developed a React-based front-end and a Python backend with FastAPI, leveraging Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, adhering to clean code practices. Version-controlled via GitLab, with CI/CD integration.
- **Tech stack: Python, FastAPI, React, LangGraph, LangChain, Postman, Docker, AWS, CDK, EC2, Lambda, GitLab, GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated RAG's Generation and Retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.

- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- **Tech stack: Python, Scikit-Learn, PyTorch, TensorFlow, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS (S3, EC2, VPC, Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker, Streamlit, Vector Databases**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Client Self-Serve Portal | TypeScript | NestJS | NextJS | FastAPI | PostgreSQL

- Self-serve portal where advertiser clients create campaigns, track delivery and download reports, with a NextJS front end, a NestJS API with typed DTOs and an OpenAPI spec, and a FastAPI service for pricing and forecasts, over seeded synthetic campaign data in PostgreSQL.
- Automates routine client requests with an LLM assistant limited to typed tool calls, shows pending actions to the internal team, and runs lint, type checks and tests in CI.

TECHNICAL SKILLSET

- **Programming:** Python, FastAPI, REST APIs, Python microservices
- **Frontend:** React(beginner), Streamlit
- **AI & Agents:** LLMs, Agentic Workflow (LangGraph), LangChain, LiteLLM, MCP, RAG, Claude Code (custom skills, agentic coding), Hugging Face Transformers
- **Databases:** PostgreSQL, MongoDB, SQL, Neo4j, Vector Databases(Qdrant, Pinecone, Azure AI Search)
- **Tools:** Git, GitLab CI/CD, Docker, Kubernetes, Postman, Scikit-Learn, PyTorch, TensorFlow, Agile
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)
- **Cloud:** Azure, AWS- S3, EC2, Lambda, SQS, Load balancer, DynamoDB, VPC, CDK

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles